PhD candidate assignment report – Mixed traffic flow

Problem and modelling assumptions:

The problem is a single-lane mixed traffic consisting of 11 human-driven vehicles (HDVs) and 4 autonomous vehicles (AVs) with pedestrian crossing at 400m, from 20s to 30s. The following assumptions are used:

- All vehicles have an acceleration range of -8 to +3m/s².
- The maximum vehicle speed is 25m/s and vehicles cannot move backwards (i.e. minimum speed is 0m/s).
- The position of each vehicle is given by its front bumper. The minimum bumper-to-bumper gap is desired to be 3m (s_{gap}).
- All vehicles have the same length of 4.5m (L).
- A 3s warning period is provided before the pedestrian enters the crossing.
- All vehicles try to stop at a safe distance of 5m from the pedestrian crossing (x_{stop} = 395m) during pedestrian crossing.
- The front-most vehicle (HDV) aims to achieve the desired speed of 25m/s.
- AVs have a desired speed of 20m/s and a desired time headway (THW) of 1.5s.

Solution approach:

A rule-based controller is used for HDVs as their behavior is already given by a car-following model. For AVs, model predictive control (MPC) is used so that they can look ahead over a prediction horizon and choose its acceleration while satisfying the speed, gap and pedestrian constraints. A 3s horizon is used to give the AV enough time to react to the leading vehicle and the approaching pedestrian without making the optimization unnecessarily large. The pedestrian warning is included for the vehicles to decide between clearing the pedestrian crossing and stopping. The active pedestrian period is handled by a stopping constraint. The simulation time step was also compared at 1, 10 and 25Hz to check the trade-off between numerical accuracy and computational cost.

Control formulation:

- **HDV: rule-based car-following model**

  For normal driving, the acceleration is given by: $a_t = \alpha(v_lead - v) + \beta(s - s*)$, where $s = x_{lead} - L - x$, is the bumper-to-bumper gap. During the pedestrian warning, the HDV calculates the deceleration needed to stop at the target: $a_{req} = -\frac{v^2}{2\,(x_{target} - x)}$. Here, x_{target} is either the stopping position behind a leader that is already stopping for the pedestrian ($x_{lead} - L - s_{gap}$) or the safe stop line (x_{stop}). If the required deceleration, a_{req} reaches the comfort-braking limit of -2m/s², the HDV enters the STOPPING mode and brakes toward the target. If the required deceleration to stop at the safe stop line exceeds the maximum available braking, the HDV checks whether it can still clear the crossing before the pedestrian arrives. If it cannot clear the crossing either, that time step is marked as infeasible, for simulation diagnostics. Once the STOPPING mode is entered, it is latched so that the vehicle continues braking toward its stopping target. When pedestrian crossing is active, the HDV still behind the crossing is latched to stop.

- **AV: MPC formulation**

  Acceleration is the control input for the AV. The MPC finds the acceleration sequence over the prediction horizon that minimizes a cost function and applies the first value from that sequence. The cost function penalizes three things: deviation from the desired speed, uncomfortable changes in acceleration (jerk), and deviation from the desired time headway. During stopping, an extra cost is added to make the AV stop close to its target position. This gives a balance between comfort, efficiency and safety. The optimization is subject to speed, acceleration and minimum-gap constraints. I used SLSQP to solve this.

  During normal driving: The controller tries to find the optimal acceleration by solving

$$\boldsymbol{a}^* = \arg\min_a J_N := w_v \sum_k (v_k - 20)^2 + w_j \sum_k jerk_k^2\,\Delta t + w_h \sum_k [\max(0, (1.5 - THW_k))]^2$$

$$\text{subject to: } -8m/s^2 \le a_k \le 3m/s^2, \quad 0m/s \le v_k \le 25m/s \quad \text{and} \quad s_k \ge s_{gap}$$

The vehicle dynamics used in the prediction are: $v_{k+1} = v_k + a_k \Delta t; \quad x_{k+1} = x_k + v_k \Delta t.$

During the warning period: There are 2 MPC problems. First, the normal MPC is solved as above. Its predicted position is then checked at the time when the pedestrian appears. If $x(20sec) \geq 400 + L$, the AV can fully clear the crossing and continues with the normal trajectory. If it cannot clear the crossing, a second MPC is solved for stopping. In this case, the desired speed is changed to 0m/s and an extra cost is added to bring the AV close to the stopping target:

$$\boldsymbol{a}^* = arg\min_a J_{stop} := w_v \sum_k {v_k}^2 + w_j \sum_k jerk_k^2 \Delta t + w_h \sum_k [\max(0, 1.5 - THW_k)]^2 + w_s \sum_k (x_k - x_{target})^2$$

Subject to the earlier constraints and an extra one: $x(20sec) \leq x_{stop}$

The stopping target is, $x_{target} = \min(x_{lead} - L - s_{gap},\ x_{stop})$; $jerk_k = \frac{a_k - a_{k-1}}{\Delta t}$ and $THW_k = \frac{s_k}{v_k}$

So, the AV aims to stop either behind its leader or at the safe stop position, whichever is closer.

When the pedestrian is active: The AV no longer decides whether to cross. It solves the stopping MPC alone. In addition, the predicted position must satisfy: $x(t_k) \leq x_{stop}; \quad 20s \leq t_k \leq 30s$

This keeps the AV before the safe stop position while the pedestrian is present.

Results and discussions:

The time step affects safety, accuracy, comfort, and computational cost. Smaller steps give more accurate updates but require more MPC evaluations. Larger time steps are cheaper but can miss important changes in speed and position. Increasing the frequency from 10 to 25Hz greatly increased the number of solves but changed the vehicle trajectories only slightly. So, a 0.1s time step (10Hz) was selected for the reported simulation.

The 60s simulation included a pedestrian warning from 17-20s and an active crossing from 20-30s. No infeasible HDV events occurred, indicating that the HDV rules handled the pedestrian event well. The AV MPC success rate was 99.29% (2,387 of 2,404 solves). During the early warning phase, some AVs briefly entered the dilemma state because neither safe stopping nor crossing was feasible due to failed solves. They then switched to stopping mode. These failed solves were handled using a fallback braking of -8m/s².

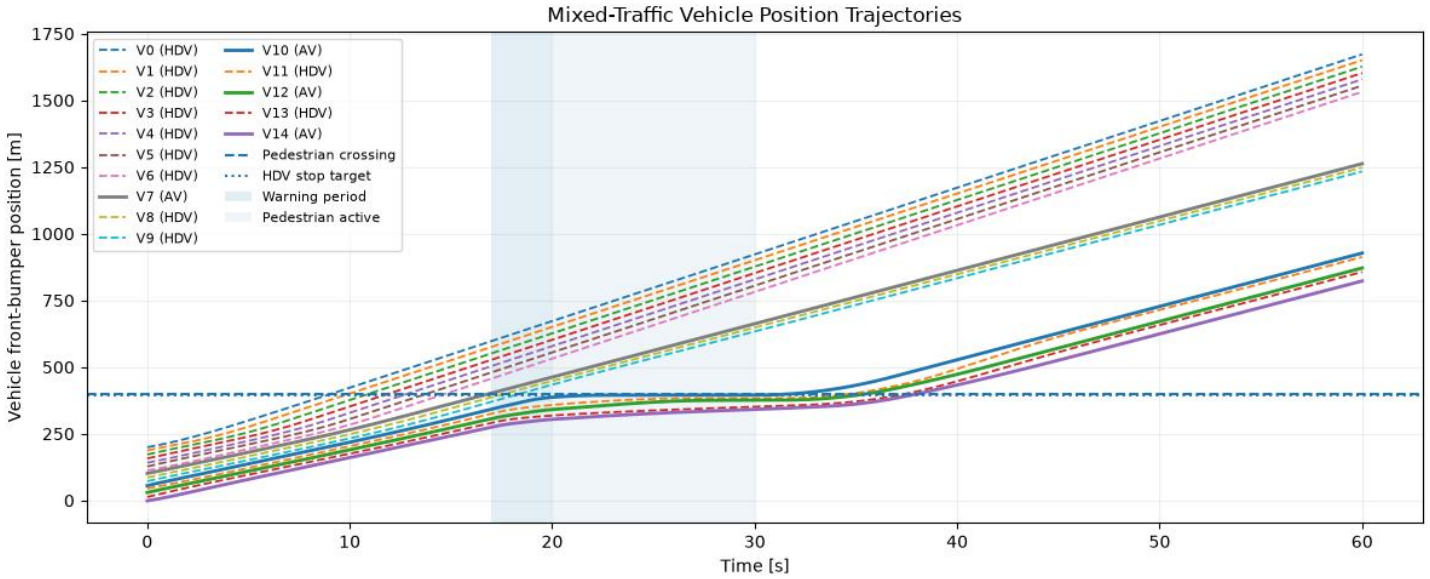
The minimum vehicle gap was 3m for all vehicles, meeting the required limit. V14 had the lowest THW of 0.287s at 37.8s, with a 4.644m gap and a speed of 16.2m/s. This short headway occurred during traffic recovery after the pedestrian event. Thus, the desired 1.5s headway was not always met because it is a soft cost in the MPC objective, rather than a hard constraint.

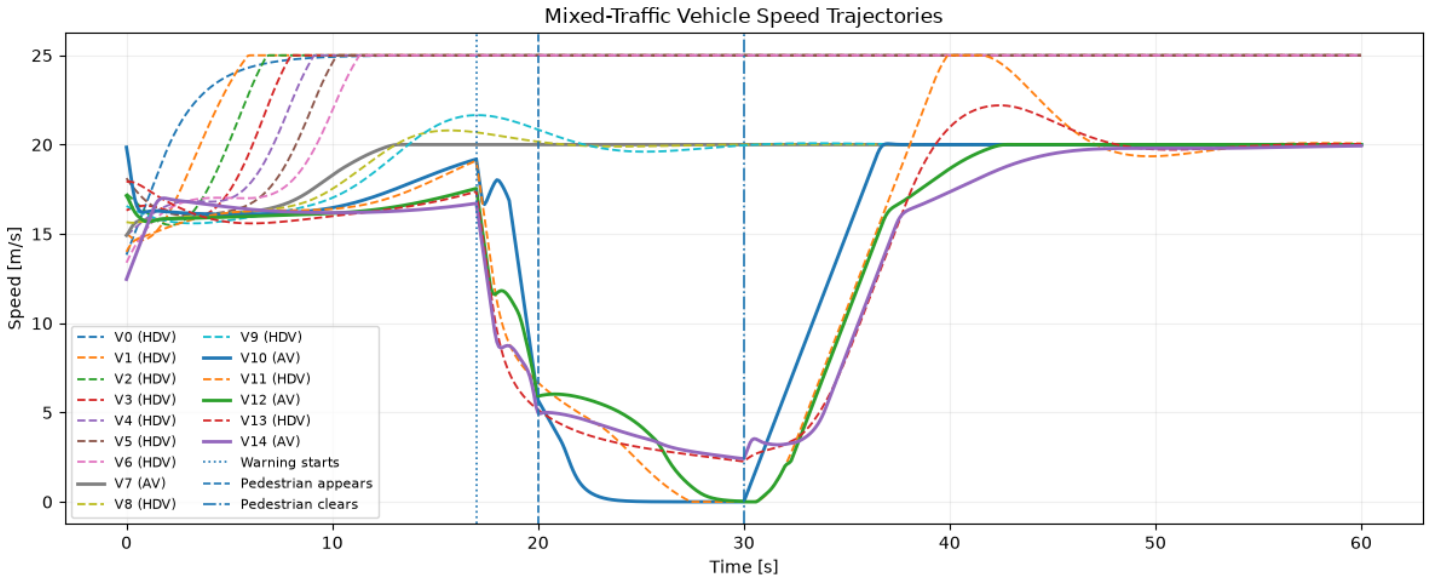
The position and velocity plots show that vehicles slow down during the warning period. From 20-30s, AV10, AV12, and AV14 stop before the crossing, while AV7 clears it before the pedestrian arrives. After 30s, they accelerate and return to their normal speeds.

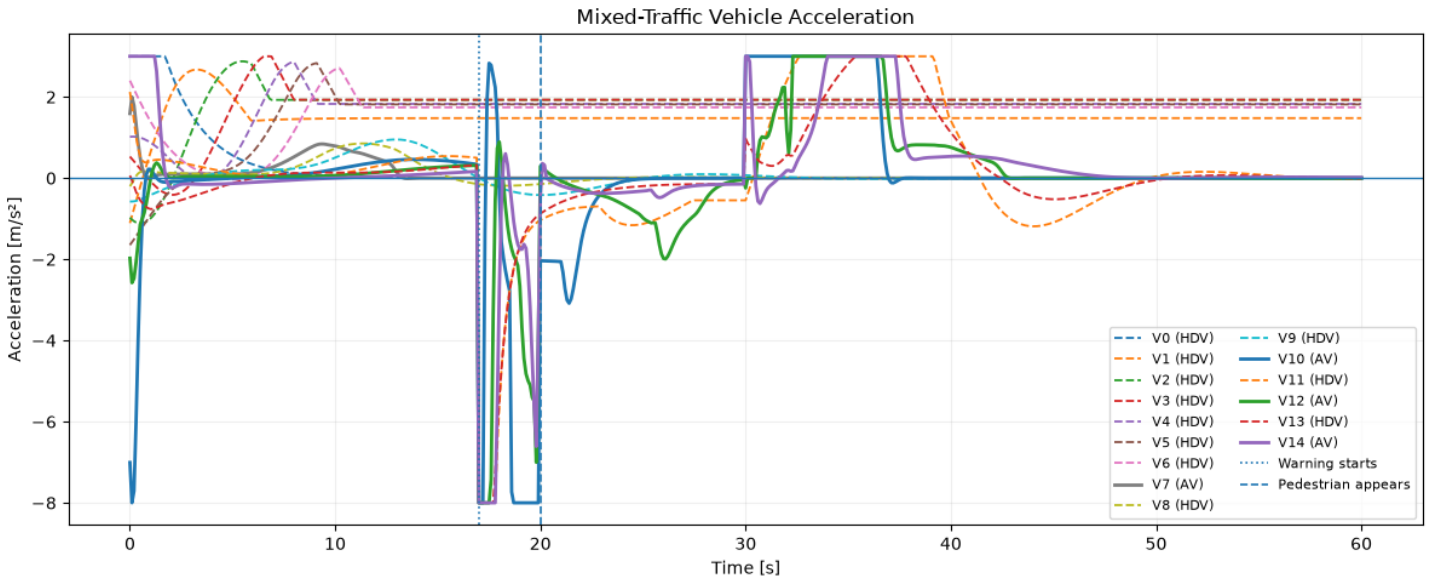
The acceleration plot shows strong braking during the pedestrian event to prioritize safety over comfort. However, in dilemma mode, the fallback creates a sudden change to -8m/s², causing high jerk. The maximum absolute jerk was about 50m/s³. Although jerks are penalized in the MPC cost, the hard pedestrian constraint can still require rapid control actions.

Overall, the pedestrian event causes brief MPC difficulties during the warning phase. The emergency fallback helps respond to situations in which safe stopping cannot be guaranteed by the optimizer. However, this fallback can reduce ride comfort by producing high jerks. Future work should reduce solver failures and make the fallback smoother without weakening pedestrian safety.

Usage of AI: I used ChatGPT for coding assistance, debugging, checking control logic, and grammar editing. The modelling assumptions, controller structure, interpretation of results, and conclusions were done by me.


Mixed-Traffic Vehicle Position Trajectories
Vehicle front-bumper position [m]
Time [s]
V0 (HDV)
V1 (HDV)
V2 (HDV)
V3 (HDV)
V4 (HDV)
V5 (HDV)
V6 (HDV)
V7 (AV)
V8 (HDV)
V9 (HDV)
V10 (AV)
V11 (HDV)
V12 (AV)
V13 (HDV)
V14 (AV)
Pedestrian crossing
HDV stop target
Warning period
Pedestrian active


Mixed-Traffic Vehicle Speed Trajectories
Speed [m/s]
Time [s]
V0 (HDV)
V1 (HDV)
V2 (HDV)
V3 (HDV)
V4 (HDV)
V5 (HDV)
V6 (HDV)
V7 (AV)
V8 (HDV)
V9 (HDV)
V10 (AV)
V11 (HDV)
V12 (AV)
V13 (HDV)
V14 (AV)
Warning starts
Pedestrian appears
Pedestrian clears


Mixed-Traffic Vehicle Acceleration
Acceleration [m/s²]
Time [s]
V0 (HDV)
V1 (HDV)
V2 (HDV)
V3 (HDV)
V4 (HDV)
V5 (HDV)
V6 (HDV)
V7 (AV)
V8 (HDV)
V9 (HDV)
V10 (AV)
V11 (HDV)
V12 (AV)
V13 (HDV)
V14 (AV)
Warning starts
Pedestrian appears